

Siddharth Banerjee

Linkedin: Siddharth Banerjee
Github: github.com/Sibidine

Email: siddharth7775banerjee@gmail.com
Mobile: +91-90514 58230

EDUCATION

- Indian Institute of Technology, Kanpur** M.Tech- Computer Science and Engineering, CPI: 8.9
July 2025- Present
- Jadavpur University, Kolkata** B.E. - Information Technology, CGPA: 8.6
December 2021 - July 2025
- St. Xavier's Collegiate School, Kolkata**
2021
Secured 98% in ICSE, 95% in ISC

TECHNICAL SKILLS

- Languages** Python, C, C++, SQL, Bash
- Frameworks** FastAPI, PyTorch, TensorFlow, Keras
- Tools** Git, PostgreSQL, LaTeX
- Platforms** Linux, Windows

COURSES

Database Management Systems	Design and Analysis of Algorithms
Distributed Computing	Introduction to Machine Learning
Operating Systems	Computational Genomics
Optimisation Techniques in Engineering Design	

WORK EXPERIENCE

- Jadavpur University** On-Site
Undergraduate Student Researcher
Jul 2023-Jul 2024
 - Project Report**
 - Working on an Instance Segmentation and Object Detection project under Dr. Sarbani Roy.
 - Utilised models such as Yolov8 to detect unoccupied and total available area on footpaths.
 - An annotated dataset of 226 images was used, with the final model having mean average precision bounding box precision of 0.985
 - Technologies used-** Python, Yolov8, LaTeX
- ITC Infotech** On-site
Project Intern
Jan 2023 - Mar 2023
 - Project Document — Letter of Completion**
 - Worked on a database administration project based on a World Bank Dataset.
 - Utilized Data Transformation using SQL commands and stored procedures to normalise the dataset of about 800,000 rows on educational surveys into logically grouped and analysable tables, according to topics provided in the metadata.
 - Cut down on general query time from 20 seconds to under 1.
 - Technologies used-** PostgreSQL, pgAdmin, PLpgSQL

PROJECTS

- Delay Recovery by Inverting a XOR Arbiter PUF.** (PG)
 - Developed a method to take a 1089-dimensional linear model corresponding to a XOR ar- biter PUF and produce 256 non-negative delays that generate the same linear model, as part of a course project under Prof. Purushottam Kar.
 - A solution to the XOR Arbiter PUF inversion problem to recover delays was developed using Singular Value Decomposition.
 - The Euclidean distance between the original model and the one reconstructed from delays was $5.1621e-16$.
- Generative Classification and Reconstruction.** (PG)
 - Worked with a team of 6 to train a generative multi-classification model where each class-conditional distribution is modeled as a mixture of K Gaussians, as part of a course project under Prof. Purushottam Kar.
 - Detailed calculations on how inference could be performed with GMM- based generative multi-classification model on a new test point were derived.
 - The formulation was extended to perform generative classification on the MNIST dataset using GMMs for each digit, alongside reconstruction in the presence of missing pixels.
- Inference of Spatial Metacells by integrating Single-Cell and Spatial Datasets.** (PG)
 - Led a team of 4 to develop metacells containing single-cell and spatial information as part of a course project under Prof. Hamim Zafar.
 - Utilized a conditional autoencoder to generate a co-embedding of spatial and single-cell data in the latent space, with spatial information being represented as a shifted covariance matrix.

- This combined information was represented in multiple ways to be passed to the metacell algorithm to generate metacells containing both single-cell and spatial context.
- **Imputation in Spatial Transcriptomics Data.** (PG)
 - Performed spatial imputation on a single-cell, single-nucleus, and spatial RNA Sequencing dataset of the Human Liver as part of a course project under Prof. Hamim Zafar.
 - Utilized Seurat's *TransferData* imputation method to perform spatial imputation.
 - Obtained an average Pearson correlation coefficient score of 0.65 and average SSIM of 0.40.
- **Self-Citation Detection Using Author Name Disambiguation.** (UG)
 - Developed an author name disambiguation-based approach to detecting self-citation in a list of references under Dr. Kartick Chandra Mondal.
 - Utilized an ensemble of machine learning and string processing algorithms to achieve an accuracy of 99.53%.
 - The paper based on the above got accepted into the Seventh International Conference on Computational Intelligence in Communications and Business Analytics (CICBA), 2025.
- **KnapSackBench.** (UG)
 - Utilized the 0-1 Knapsack Problem to benchmark the performance of various non-traditional optimization algorithms, such as the Genetic Algorithm, Simulated Annealing and Particle Swarm Optimization, among others.
 - Improved performance by at least 45% over traditional Dynamic Programming approaches.
 - Results and metrics of evaluation are made available via FastAPI.(Code)

HONOURS AND AWARDS

- GATE CS All India Rank of **88** out of 124,000 candidates. -2024
- WBJEE General Merit rank of **259** out of 92,000 candidates.- 2021
- Letter of Recommendation from the Chief Minister for being in the state **top 10** in the board examinations. - 2019

VOLUNTEER EXPERIENCE

- **Google Developer Students Club JU**

Content Lead

Aug 2023-July 2024

Involved in setting problems and creating editorials for algorithmic contests with 200+ participants, anchoring events, and managing other facets of the club.

- **ACM JU student chapter**

Membership Chair

Aug 2022-July 2024

Responsible for setting up and organising seminars and hackathons of 100+ participants, as well as aiding new members in joining the student chapter.